

1. Questions

Read the following information carefully and answer the questions based on it.

The funnel chart given below shows the total number of students who enquired for the admission in two colleges P and Q. It also shows the percentage of the number of students who appeared in the entrance test out of the total number of students who enquired for the admission, the ratio of the number of students who passed and failed in the entrance test and the number of students who took admission in each college.

	College P	College Q
Total number of students who enquired for the admission	200X	4000
Percentage of the number of students who appeared in the entrance test	90%	Y%
Ratio of the number of students who passed and failed in the entrance test	W:2Z	3:2
Number of students who took admission	240W	800Z

Note:

The total number of students who enquired for the admission = the number of students who appeared in the entrance test + the number of students who did not appear in the entrance test

The total number of students who appeared in the entrance test = the number of students who passed in the entrance test + the number of students who failed in the entrance test

The total number of students who passed in the entrance test = the number of students who took admission + the number of students who did not take admission

The number of students who appeared in the entrance test from college P is 300 less than college Q.

The number of students who did not appear in the entrance test from college P is 30% of the number of students who did not appear in the entrance test from college Q.

The number of students who failed in the entrance test from colleges P and Q are the same.

In college P, if 40% of the total number of students who passed in the entrance test are girls and $\frac{2}{3}^{\text{rd}}$ of the total number of students who did not take admission are boys, then find the difference between the number of boys and girls who took admission in college P.

- 200
- 100

- c. 300
- d. 400
- e. None of these

2. Questions

In college Q, if the number of students who took admission in post-graduation is 50% more than the difference between the number of students who took admission in graduation and post-graduation, then find the number of students who took admission in graduation in college Q. (graduation > post-graduation)

- a. 600
- b. 1000
- c. 400
- d. 1200
- e. None of these

3. Questions

Which of the following equation(s) is/are correct?

Equation I: $(X/W) = Z$

Equation II: $(WX) = Y$

Equation III: $(X-Z) = W$

- a. Only II follow
- b. Both I and III follow
- c. Both II and III follow
- d. All I, II, and III follow
- e. None follow

4. Questions

@QuizByMickey

In another college R, if the number of students who did not appear in the entrance test is 40% less than that of college Q and the number of students who appeared in the entrance test is 1800, then find the total number of students who enquired for the admission in college R.

- a. 2000
- b. 1200
- c. 2400
- d. 1600

e. 1800

5. Questions

In college Q, if the total number of students who enquired for the admission is increased by 400 and the number of students who appeared in the entrance test is increased by ___[?_]___%, then the number of students who did not appear in the entrance test in college Q will be 1100.

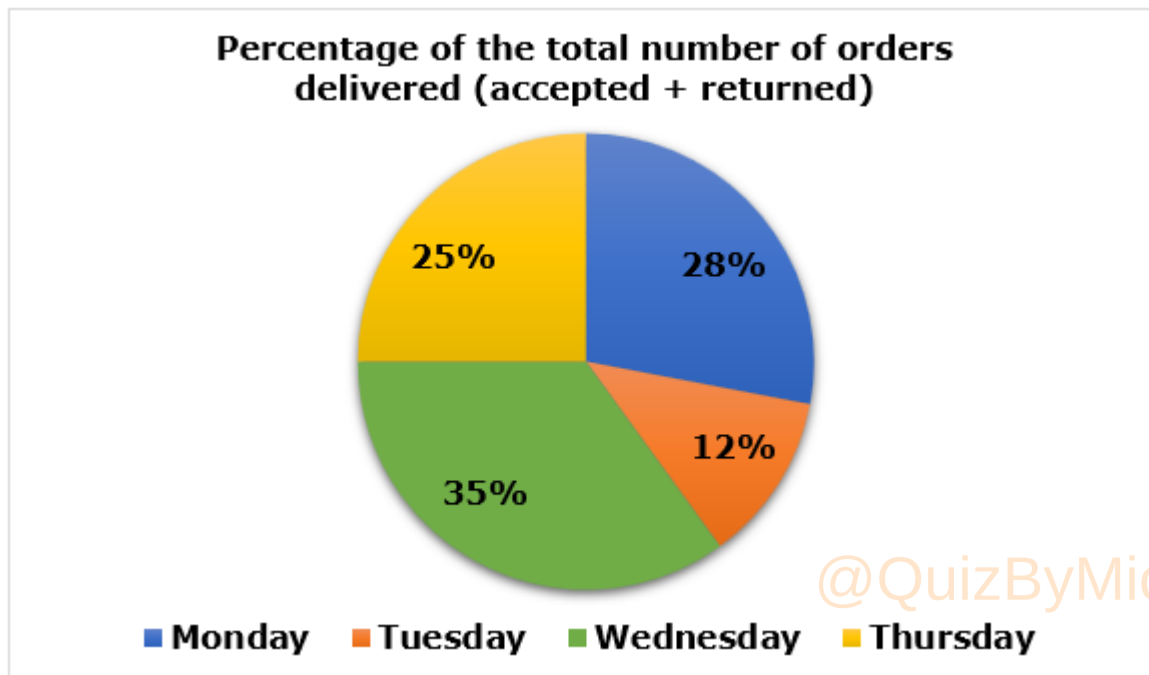
What should come in place of the question mark?

- a. 25
- b. 10
- c. 50
- d. 35
- e. 15

6. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of orders delivered (accepted + returned) by a shop on four different days Monday, Tuesday, Wednesday, and Thursday.



Note:

The total number of orders placed = the number of orders delivered + the number of non-delivered orders

The total number of orders delivered = the number of orders accepted + the number of orders returned

The total number of orders placed on Monday, Tuesday, Wednesday, and Thursday is 140, 64, 200, and 120 respectively.

The number of orders accepted on Tuesday is 30, which is 47 less than the previous day.

The total number of orders returned on Monday and Tuesday is 53.

The ratio of the number of orders accepted on Wednesday and Thursday is 8:5.

The difference between the number of orders returned on Wednesday and Thursday is 5. (Wednesday < Thursday)

On Wednesday, out of the total number of orders accepted, some orders remained unsold, while the rest were sold. If the ratio of the number of orders sold to males and females is 2:1 and the number of orders sold to females is 8 more than the number of orders unsold, then find the total number of orders sold (males + females) on Monday.

- a. 60
- b. 96
- c. 72
- d. 84
- e. None of these

7. Questions

If the number of non-delivered orders on Thursday are further delivered on the next day and the total number of orders placed on Friday is 80, out of which $\frac{1}{5}^{\text{th}}$ are non-delivered orders, then find the number of orders delivered on Friday (excluding non-delivered orders of the previous day).

- a. 52
- b. 36
- c. 44
- d. Cannot be determined
- e. None of these

8. Questions

Out of the total number of items delivered on Monday, $\frac{1}{4}^{\text{th}}$ are grocery items, while the rest are electronic items. If the ratio of the number of electronic and grocery items returned on Monday is 3:2, then find what percent of the total number of orders accepted on Monday are electronic items?

- a. 57.14%
- b. 85.71%
- c. 71.42%
- d. 81.81%
- e. None of these

9. Questions

On Tuesday, out of the total number of orders accepted, some orders are sold, while the rest remained unsold. If the total cost of all sold orders is Rs. 2000 more than the unsold orders and the price of each order is Rs. 200, then find the ratio of the number of orders sold and unsold on Tuesday.

- a. 2:1
- b. 3:2
- c. 4:1
- d. Cannot be determined
- e. None of these

10. Questions

On Saturday, if the number of delivered orders is twice of Thursday and 16.67% of the total number of orders placed on Saturday, are non-delivered orders, then find the total number of orders placed on Saturday.

- a. 660
- b. 300
- c. 420
- d. 240
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

A shop manufacture two types of furniture tables and chairs in three different years 2022, 2023 and 2024.

2022: The total number of furniture (tables + chairs) manufactured is 500, out of which $\frac{2}{5}^{\text{th}}$ are chairs. The number of tables unsold is 40% of the number of chairs sold. The ratio of the number of tables sold and the number of chairs unsold is 24:5.

2023: The total number of tables manufactured is 20 less than the previous year. 30% of the total number of furniture (tables + chairs) manufactured are chairs. The number of tables unsold is 60% less than the number of chairs sold. The number of tables sold is 12 times of the number of chairs unsold.

2024: The total number of chairs manufactured is 50, which is $\frac{1}{3}^{\text{rd}}$ of the total number of furniture tables manufactured. The ratio of the number of tables sold and the number of chairs unsold is 9:2. The number of chairs sold is half of the number of tables unsold.

Note:

The total number of furniture manufactured = the number of tables manufactured + the number of chairs manufactured

The total number of tables manufactured = the number of tables sold + the number of tables unsold

The total number of chairs manufactured = the number of chairs sold + the number of chairs unsold

In 2022, if [X]% of the total number of tables manufactured are defective, and the number of non-defective tables unsold will be $\frac{1}{8}$ th of the number of non-defective tables sold. Find X.(Only non-defective tables can be sold)

- a. 20
- b. 30
- c. 40
- d. 10
- e. 90

12. Questions

In 2023, if the price of each wooden and plastic chair is Rs. 200 and Rs. 100 respectively and the total revenue generated from the sale of all chairs (wooden + plastic) is Rs. 16000, then find the number of wooden chairs sold by the shop in 2023.

- a. 60
- b. 40
- c. 75
- d. 25
- e. None of these

13. Questions

In 2021, if the number of furniture sold is 90 more than 2022 and the number of furniture sold to males is 360, then find the total number of furniture manufactured in 2021, if the number of furniture sold to females is equals to the number of furniture unsold in 2021.

- a. 480
- b. 600
- c. 360
- d. Cannot be determined
- e. None of these

@QuizByMickey

14. Questions

In 2024, if [Y]% of the total number of furniture (tables + chairs) unsold are further sold, then the ratio of the number of furniture sold and unsold will be 7:3. Find the value of [2Y].

- a. 42

- b. 34
- c. 50
- d. 28
- e. None of these

15. Questions

In 2022, if __[?]% of the total number of chairs sold is further returned, then the number of chairs unsold will be 85 less than the initial number of chairs sold. Find the value of question mark. (The total number of chairs manufactured in 2022 remained unchanged)

- 18
- 10
- 24
- 32
- None of these

16. Questions

The following question has two series as series I and series II. Find the pattern of series I and series II in each question and answer accordingly.

Series I and II have one missing term as [P] and [Q] respectively. Both series follow different patterns.

Series I: 33, 11, 66, __[P]__, 132, 44

Series II: 14, 34, 9, 39, 4, __[Q]__

Which of the following is/are correct?

I: $2P = Q$

II: [Q-22] = P

III: $[Q/P] = 2$

- Only I follow
- Both I and II follow
- Both II and III follow
- All I, II and III follow
- None follow

17. Questions

Series I has one wrong term and Series II has two missing terms [M] and [N]. Both series follow the

same pattern.

Series I: 3, 5, 13, 48, 241, 1441

Series II: 8, 15, 43, __[M]__, __[N]__, 5041

If the correct term of series I is [K], then which of the following is a perfect square?

- a. K
- b. M
- c. N
- d. All of the above
- e. None of these

18. Questions

Series I and II have one wrong term. [X] and [Y] are correct terms that will replace the wrong terms of series I and II respectively. Both series follow the same pattern.

Series I: 10, 14, 23, 39, 97, 218

Series II: 30, 34, 43, 68, 104, 238

Which of the following is/are definitely true?

I: Both [X] and [Y] are exactly divisible by 5.

II: The sum of the digits of [Y] is 9.

III: [X] is 1 less than the square of 7.

- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

19. Questions

Series II follows the same pattern as Series I.

Series I: 28, [A], [B], [C], 657, 3284

Series II: 16, ____, ____, ____, ____, __[?]__

A and B are two-digit natural numbers.

A is a perfect cube of the smallest odd prime number.

B is an exact multiple of both 5 and 11.

C is 5 less than the square of the second lowest two-digit prime number.

@QuizByMickey

What should come in place of the question mark?

- a. 1755
- b. 1844
- c. 1936
- d. 1562
- e. None of these

20. Questions

Series I has one missing term and series II has one wrong term. Both series follow different patterns.

Series I: 4, 8, 2, 16, __[K]__, 32

Series II: __[K]__, 13, 37, 86, 181, 373

Find the correct term that will replace the wrong term of series II.

- a. 71
- b. 66
- c. 85
- d. 52
- e. None of these

Explanations:

1. Questions

The total number of students who enquired for the admission in college P = $200X$

The number of students who appeared in the entrance test in college P = 90% of $200X = 180X$

The number of students who did not appear in the entrance test in college P = $(200X - 180X) = 20X$

Also,

@QuizByMickey

The total number of students who enquired for the admission in college Q = 4000

The number of students who appeared in the entrance test in college Q = $Y\%$ of 4000 = $40Y$

The number of students who did not appear in the entrance test in college Q = $(4000 - 40Y)$

Now,

$$40Y - 180X = 300$$

$$2Y - 9X = 15 \dots\dots\dots(i)$$

And,

$$20X/(4000-40Y) = 3/10$$

$$200X = 12000 - 120Y$$

$$120Y + 200X = 12000$$

$$3Y + 5X = 300 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$6Y - 27X = 45$$

$$6Y + 10X = 600$$

$$37X = 555$$

$$X = 15$$

Now, put the value of X in equation (i), we get,

$$2Y - 9X = 15$$

$$2Y - (9 \times 15) = 15$$

$$2Y = 15 + 135$$

$$2Y = 150$$

$$Y = 75$$

College Q

The total number of students who enquired for the admission = 4000

The number of students who appeared in the entrance test = Y% of 4000 = 75% of 4000 = 3000

The number of students who did not appear in the entrance test = (4000-3000) = 1000

The number of students who passed in the entrance test = 3/5 of 3000 = 1800

The number of students who failed in the entrance test = 2/5 of 3000 = 1200

College P

The total number of students who enquired for the admission = 200X = (200 × 15) = 3000

The number of students who appeared in the entrance test = 90% of 3000 = 2700

The number of students who did not appear in the entrance test = (3000-2700) = 300

The number of students who failed in the entrance test = 1200

The number of students who passed in the entrance test = (2700-1200) = 1500

So, the ratio of the number of students who passed and failed in the entrance test = 1500:1200 = 5:4.

Now,

$$W:2Z = 5:4$$

So, $W = 5$

And, $2Z = 4$

$Z = 2$

Therefore,

The number of students who took admission from college P = $240W = (240 \times 5) = 1200$

The number of students who took admission from college Q = $800Z = (800 \times 2) = 1600$

The number of students who did not take admission from college P = $(1500 - 1200) = 300$

The number of students who did not take admission from college Q = $(1800 - 1600) = 200$

College	The total number of students who enquired for the admission	The number of students who appeared in the entrance test	The number of students who did not appear in the entrance test	The number of students who passed in the entrance test	The number of students who failed in the entrance test	The number of students who took admission	The number of students who did not take admission
P	3000	2700	300	1500	1200	1200	300
Q	4000	3000	1000	1800	1200	1600	200
Total	7000	5700	1300	3300	2400	2800	500

Answer: A

Now, according to the question,

College P

The total number of students who passed in the entrance test = 1500

The number of girls who passed in the entrance test = 40% of 1500 = 600

So, the number of boys who passed in the entrance test = $(1500 - 600) = 900$

The total number of students who did not take admission = 300

The number of boys who did not take admission = $\frac{2}{3}$ of 300 = 200

So, the number of girls who did not take admission = $(300 - 200) = 100$

The number of boys who took admission = $(900 - 200) = 700$

And, the number of girls who took admission = $(600 - 100) = 500$

Therefore, the required difference = $(700-500) = 200$.

Hence, the correct answer is option A.

2. Questions

The total number of students who enquired for the admission in college P = $200X$

The number of students who appeared in the entrance test in college P = 90% of $200X = 180X$

The number of students who did not appear in the entrance test in college P = $(200X-180X) = 20X$

Also,

The total number of students who enquired for the admission in college Q = 4000

The number of students who appeared in the entrance test in college Q = $Y\%$ of $4000 = 40Y$

The number of students who did not appear in the entrance test in college Q = $(4000-40Y)$

Now,

$$40Y - 180X = 300$$

$$2Y - 9X = 15 \dots\dots\dots(i)$$

And,

$$20X/(4000-40Y) = 3/10$$

$$200X = 12000 - 120Y$$

$$120Y + 200X = 12000$$

$$3Y + 5X = 300 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$6Y - 27X = 45$$

$$6Y + 10X = 600$$

$$37X = 555$$

$$X = 15$$

Now, put the value of X in equation (i), we get,

$$2Y - 9X = 15$$

$$2Y - (9 \times 15) = 15$$

$$2Y = 15 + 135$$

$$2Y = 150$$

$$Y = 75$$

College Q

The total number of students who enquired for the admission = 4000

@QuizByMickey

The number of students who appeared in the entrance test = $Y\%$ of 4000 = 75% of 4000 = 3000

The number of students who did not appear in the entrance test = $(4000-3000) = 1000$

The number of students who passed in the entrance test = $\frac{3}{5}$ of 3000 = 1800

The number of students who failed in the entrance test = $\frac{2}{5}$ of 3000 = 1200

College P

The total number of students who enquired for the admission = $200X = (200 \times 15) = 3000$

The number of students who appeared in the entrance test = 90% of 3000 = 2700

The number of students who did not appear in the entrance test = $(3000-2700) = 300$

The number of students who failed in the entrance test = 1200

The number of students who passed in the entrance test = $(2700-1200) = 1500$

So, the ratio of the number of students who passed and failed in the entrance test = $1500:1200 = 5:4$.

Now,

$$W:2Z = 5:4$$

$$\text{So, } W = 5$$

$$\text{And, } 2Z = 4$$

$$Z = 2$$

Therefore,

The number of students who took admission from college P = $240W = (240 \times 5) = 1200$

The number of students who took admission from college Q = $800Z = (800 \times 2) = 1600$

The number of students who did not take admission from college P = $(1500-1200) = 300$

The number of students who did not take admission from college Q = $(1800-1600) = 200$

@QuizByMickey

College	The total number of students who enquired for the admission	The number of students who appeared in the entrance test	The number of students who did not appear in the entrance test	The number of students who passed in the entrance test	The number of students who failed in the entrance test	The number of students who took admission	The number of students who did not take admission
P	3000	2700	300	1500	1200	1200	300
Q	4000	3000	1000	1800	1200	1600	200
Total	7000	5700	1300	3300	2400	2800	500

Answer: B

College Q

The total number of students who took admission = 1600

The ratio of the number of students who took admission in post-graduation and the difference between the number of students who took admission in graduation and post-graduation = 150:100 = 3:2. Let it be 3k and 2k respectively.

So, the number of students who took admission in graduation = (3k+2k) = 5k

Now,

$$5k + 3k = 1600$$

$$8k = 1600$$

$$k = 200$$

Therefore, the number of students who took admission in graduation = 5k = (5×200) = 1000

Hence, the correct answer is option B

@QuizByMickey

3. Questions

The total number of students who enquired for the admission in college P = 200X

The number of students who appeared in the entrance test in college P = 90% of 200X = 180X

The number of students who did not appear in the entrance test in college P = (200X-180X) = 20X

Also,

The total number of students who enquired for the admission in college Q = 4000

The number of students who appeared in the entrance test in college Q = Y% of 4000 = 40Y

The number of students who did not appear in the entrance test in college Q = $(4000 - 40Y)$

Now,

$$40Y - 180X = 300$$

$$2Y - 9X = 15 \dots\dots\dots(i)$$

And,

$$20X / (4000 - 40Y) = 3/10$$

$$200X = 12000 - 120Y$$

$$120Y + 200X = 12000$$

$$3Y + 5X = 300 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$6Y - 27X = 45$$

$$6Y + 10X = 600$$

$$37X = 555$$

$$X = 15$$

Now, put the value of X in equation (i), we get,

$$2Y - 9X = 15$$

$$2Y - (9 \times 15) = 15$$

$$2Y = 15 + 135$$

$$2Y = 150$$

$$Y = 75$$

College Q

The total number of students who enquired for the admission = 4000

The number of students who appeared in the entrance test = $Y\%$ of 4000 = 75% of 4000 = 3000

The number of students who did not appear in the entrance test = $(4000 - 3000) = 1000$

The number of students who passed in the entrance test = $3/5$ of 3000 = 1800

The number of students who failed in the entrance test = $2/5$ of 3000 = 1200

College P

The total number of students who enquired for the admission = $200X = (200 \times 15) = 3000$

The number of students who appeared in the entrance test = 90% of 3000 = 2700

The number of students who did not appear in the entrance test = $(3000 - 2700) = 300$

The number of students who failed in the entrance test = 1200

The number of students who passed in the entrance test = $(2700-1200) = 1500$

So, the ratio of the number of students who passed and failed in the entrance test = $1500:1200 = 5:4$.

Now,

$$W:2Z = 5:4$$

$$\text{So, } W = 5$$

$$\text{And, } 2Z = 4$$

$$Z = 2$$

Therefore,

$$\text{The number of students who took admission from college P} = 240W = (240 \times 5) = 1200$$

$$\text{The number of students who took admission from college Q} = 800Z = (800 \times 2) = 1600$$

$$\text{The number of students who did not take admission from college P} = (1500-1200) = 300$$

$$\text{The number of students who did not take admission from college Q} = (1800-1600) = 200$$

College	The total number of students who enquired for the admission	The number of students who appeared in the entrance test	The number of students who did not appear in the entrance test	The number of students who passed in the entrance test	The number of students who failed in the entrance test	The number of students who took admission	The number of students who did not take admission
P	3000	2700	300	1500	1200	1200	300
Q	4000	3000	1000	1800	1200	1600	200
Total	7000	5700	1300	3300	2400	2800	500

Answer: A

Equation I: $(X/W) = Z$

$$(15/5) = 3$$

$$Z = 2$$

So, equation I is not correct.

Equation II: $(WX) = Y$

$$(WX) = (5 \times 15) = 75$$

@QuizByMickey

$$Y = 75$$

So, equation II is correct.

Equation III: $(X-Z) = W$

$$(X-Z) = (15-2) = 13$$

So, equation III is not correct.

Therefore, only II follows.

Hence, the correct answer is option A.

4. Questions

The total number of students who enquired for the admission in college P = $200X$

The number of students who appeared in the entrance test in college P = 90% of $200X = 180X$

The number of students who did not appear in the entrance test in college P = $(200X - 180X) = 20X$

Also,

The total number of students who enquired for the admission in college Q = 4000

The number of students who appeared in the entrance test in college Q = Y% of 4000 = $40Y$

The number of students who did not appear in the entrance test in college Q = $(4000 - 40Y)$

Now,

$$40Y - 180X = 300$$

$$2Y - 9X = 15 \dots\dots\dots(i)$$

And,

$$20X / (4000 - 40Y) = 3/10$$

$$200X = 12000 - 120Y$$

$$120Y + 200X = 12000$$

$$3Y + 5X = 300 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$6Y - 27X = 45$$

$$6Y + 10X = 600$$

$$37X = 555$$

$$X = 15$$

Now, put the value of X in equation (i), we get,

$$2Y - 9X = 15$$

$$2Y - (9 \times 15) = 15$$

@QuizByMickey

$$2Y = 15 + 135$$

$$2Y = 150$$

$$Y = 75$$

College Q

The total number of students who enquired for the admission = 4000

The number of students who appeared in the entrance test = $Y\%$ of 4000 = 75% of 4000 = 3000

The number of students who did not appear in the entrance test = $(4000 - 3000) = 1000$

The number of students who passed in the entrance test = $\frac{3}{5}$ of 3000 = 1800

The number of students who failed in the entrance test = $\frac{2}{5}$ of 3000 = 1200

College P

The total number of students who enquired for the admission = $200X = (200 \times 15) = 3000$

The number of students who appeared in the entrance test = 90% of 3000 = 2700

The number of students who did not appear in the entrance test = $(3000 - 2700) = 300$

The number of students who failed in the entrance test = 1200

The number of students who passed in the entrance test = $(2700 - 1200) = 1500$

So, the ratio of the number of students who passed and failed in the entrance test = $1500:1200 = 5:4$.

Now,

$$W:2Z = 5:4$$

$$\text{So, } W = 5$$

$$\text{And, } 2Z = 4$$

$$Z = 2$$

Therefore,

The number of students who took admission from college P = $240W = (240 \times 5) = 1200$

The number of students who took admission from college Q = $800Z = (800 \times 2) = 1600$

The number of students who did not take admission from college P = $(1500 - 1200) = 300$

The number of students who did not take admission from college Q = $(1800 - 1600) = 200$

College	The total number of students who enquired for the admission	The number of students who appeared in the entrance test	The number of students who did not appear in the entrance test	The number of students who passed in the entrance test	The number of students who failed in the entrance test	The number of students who took admission	The number of students who did not take admission
P	3000	2700	300	1500	1200	1200	300
Q	4000	3000	1000	1800	1200	1600	200
Total	7000	5700	1300	3300	2400	2800	500

Answer: C

College R

The number of students who did not appear in the entrance test = 60% of 1000 = 600

The number of students who appeared in the entrance test = 1800

Therefore, the total number of students who enquired for the admission = (1800+600) = 2400.

Hence, the correct answer is option C.

5. Questions

The total number of students who enquired for the admission in college P = 200X

The number of students who appeared in the entrance test in college P = 90% of 200X = 180X

The number of students who did not appear in the entrance test in college P = (200X-180X) = 20X

Also,

The total number of students who enquired for the admission in college Q = 4000

The number of students who appeared in the entrance test in college Q = Y% of 4000 = 40Y

The number of students who did not appear in the entrance test in college Q = (4000-40Y)

Now,

$$40Y - 180X = 300$$

$$2Y - 9X = 15 \dots\dots\dots(i)$$

And,

$$20X/(4000-40Y) = 3/10$$

$$200X = 12000 - 120Y$$

$$120Y + 200X = 12000$$

$$3Y + 5X = 300 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$6Y - 27X = 45$$

$$6Y + 10X = 600$$

$$37X = 555$$

$$X = 15$$

Now, put the value of X in equation (i), we get,

$$2Y - 9X = 15$$

$$2Y - (9 \times 15) = 15$$

$$2Y = 15 + 135$$

$$2Y = 150$$

$$Y = 75$$

College Q

The total number of students who enquired for the admission = 4000

The number of students who appeared in the entrance test = Y% of 4000 = 75% of 4000 = 3000

The number of students who did not appear in the entrance test = (4000-3000) = 1000

The number of students who passed in the entrance test = $\frac{3}{5}$ of 3000 = 1800

The number of students who failed in the entrance test = $\frac{2}{5}$ of 3000 = 1200

College P

The total number of students who enquired for the admission = $200X = (200 \times 15) = 3000$

The number of students who appeared in the entrance test = 90% of 3000 = 2700

The number of students who did not appear in the entrance test = (3000-2700) = 300

The number of students who failed in the entrance test = 1200

The number of students who passed in the entrance test = (2700-1200) = 1500

So, the ratio of the number of students who passed and failed in the entrance test = 1500:1200 = 5:4.

Now,

$$W:2Z = 5:4$$

$$\text{So, } W = 5$$

$$\text{And, } 2Z = 4$$

$$Z = 2$$

Therefore,

The number of students who took admission from college P = $240W = (240 \times 5) = 1200$

The number of students who took admission from college Q = $800Z = (800 \times 2) = 1600$

The number of students who did not take admission from college P = $(1500 - 1200) = 300$

The number of students who did not take admission from college Q = $(1800 - 1600) = 200$

College	The total number of students who enquired for the admission	The number of students who appeared in the entrance test	The number of students who did not appear in the entrance test	The number of students who passed in the entrance test	The number of students who failed in the entrance test	The number of students who took admission	The number of students who did not take admission
P	3000	2700	300	1500	1200	1200	300
Q	4000	3000	1000	1800	1200	1600	200
Total	7000	5700	1300	3300	2400	2800	500

Answer: B

College Q

The total number of students who enquired for the admission = $(4000 + 400) = 4400$

The number of students who did not appear in the entrance test = 1100

So, the number of students who appeared in the entrance test = $(4400 - 1100) = 3300$

And, the initial number of students who appeared in the entrance test = 3000

Therefore, $? = \left[\frac{(3300 - 3000)}{3000} \times 100 \right] = \left[\frac{(300)}{3000} \times 100 \right] = 10$. @QuizByMickey

Hence, the correct answer is option B.

6. Questions

Let the total number of orders delivered on all four days together be 100a.

Now,

The number of delivered on Monday = 28% of 100a = 28a

The number of orders accepted on Monday = $(30 + 47) = 77$

The number of orders returned on Monday = $(28a-77)$

Also,

The number of delivered on Tuesday = 12% of $100a = 12a$

The number of orders accepted on Tuesday = 30

The number of orders returned on Tuesday = $(12a-30)$

So,

$$(28a-77) + (12a-30) = 53$$

$$40a - 107 = 53$$

$$40a = 160$$

$$a = 4$$

Therefore, the total number of orders delivered on all four days together = $100a = (100 \times 4) = 400$.

Monday

The total number of orders placed = 140

The number of orders delivered = 28% of $400 = 112$

The number of non-delivered orders = $(140-112) = 28$

The number of orders accepted = 77

The number of orders returned = $(112-77) = 35$

Tuesday

The total number of orders placed = 64

The number of orders delivered = 12% of $400 = 48$

The number of non-delivered orders = $(64-48) = 16$

The number of orders accepted = 30

The number of orders returned = $(48-30) = 18$

Now, let the number of orders accepted on Wednesday and Thursday be $8b$ and $5b$ respectively.

Wednesday

The total number of orders placed = 200

The number of orders delivered = 35% of $400 = 140$

The number of non-delivered orders = $(200-140) = 60$

The number of orders accepted = $8b$

The number of orders returned = $(140-8b)$

Thursday

The total number of orders placed = 120

The number of orders delivered = 25% of 400 = 100

The number of non-delivered orders = (120-100) = 20

The number of orders accepted = 5b

The number of orders returned = (100-5b)

So,

$$(100-5b) - (140-8b) = 5$$

$$95 - 5b = 140 - 8b$$

$$3b = 45$$

$$b = 15$$

Therefore,

The number of orders accepted on Wednesday = $8b = (8 \times 15) = 120$

The number of orders accepted on Thursday = $5b = (5 \times 15) = 75$

The number of orders returned on Wednesday = $(140-8b) = (140-120) = 20$

The number of orders returned on Thursday = $(100-5b) = (100-75) = 25$

Day	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of orders accepted	The number of orders returned
Monday	140	112	28	77	35
Tuesday	64	48	16	30	18
Wednesday	200	140	60	120	20
Thursday	120	100	20	75	25
Total	524	400	124	302	98

Answer: B

@QuizByMickey

Now, according to the question,

Wednesday

The total number of orders accepted = 120

Let the number of orders sold to males and females be 2k and k respectively.

So, the total number of orders sold (males + females) = $(2k+k) = 3k$

And, the number of orders unsold = $(120-3k)$

Now,

$$k - (120 - 3k) = 8$$

$$4k - 120 = 8$$

$$4k = 128$$

$$k = 32$$

Therefore, the total number of orders sold (males + females) on Wednesday = $3k = (3 \times 32) = 96$.

Hence, the correct answer is option B.

7. Questions

Let the total number of orders delivered on all four days together be $100a$.

Now,

The number of delivered on Monday = 28% of $100a = 28a$

The number of orders accepted on Monday = $(30 + 47) = 77$

The number of orders returned on Monday = $(28a - 77)$

Also,

The number of delivered on Tuesday = 12% of $100a = 12a$

The number of orders accepted on Tuesday = 30

The number of orders returned on Tuesday = $(12a - 30)$

So,

$$(28a - 77) + (12a - 30) = 53$$

$$40a - 107 = 53$$

$$40a = 160$$

$$a = 4$$

Therefore, the total number of orders delivered on all four days together = $100a = (100 \times 4) = 400$.

Monday

The total number of orders placed = 140

The number of orders delivered = 28% of $400 = 112$

The number of non-delivered orders = $(140 - 112) = 28$

The number of orders accepted = 77

The number of orders returned = $(112 - 77) = 35$

Tuesday

The total number of orders placed = 64

The number of orders delivered = 12% of $400 = 48$

@QuizByMickey

The number of non-delivered orders = $(64-48) = 16$

The number of orders accepted = 30

The number of orders returned = $(48-30) = 18$

Now, let the number of orders accepted on Wednesday and Thursday be $8b$ and $5b$ respectively.

Wednesday

The total number of orders placed = 200

The number of orders delivered = 35% of 400 = 140

The number of non-delivered orders = $(200-140) = 60$

The number of orders accepted = $8b$

The number of orders returned = $(140-8b)$

Thursday

The total number of orders placed = 120

The number of orders delivered = 25% of 400 = 100

The number of non-delivered orders = $(120-100) = 20$

The number of orders accepted = $5b$

The number of orders returned = $(100-5b)$

So,

$$(100-5b) - (140-8b) = 5$$

$$95 - 5b = 140 - 8b$$

$$3b = 45$$

$$b = 15$$

Therefore,

The number of orders accepted on Wednesday = $8b = (8 \times 15) = 120$

The number of orders accepted on Thursday = $5b = (5 \times 15) = 75$

The number of orders returned on Wednesday = $(140-8b) = (140-120) = 20$

The number of orders returned on Thursday = $(100-5b) = (100-75) = 25$

@QuizByMickey

Day	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of orders accepted	The number of orders returned
Monday	140	112	28	77	35
Tuesday	64	48	16	30	18
Wednesday	200	140	60	120	20
Thursday	120	100	20	75	25
Total	524	400	124	302	98

Answer: C

The number of non-delivered orders on Thursday = 20

The total number of orders placed on Friday = 80

The number of non-delivered orders on Friday = $\frac{1}{5}$ of 80 = 16

So, the number of delivered orders on Friday = $(80-16) = 64$ (including non-delivered orders of Thursday)

Therefore, the number of delivered orders on Friday (excluding non-delivered orders of Thursday) = $(64-20) = 44$.

Hence, the correct answer is option C.

8. Questions

Let the total number of orders delivered on all four days together be 100a.

Now,

The number of delivered on Monday = 28% of 100a = 28a

The number of orders accepted on Monday = $(30+47) = 77$

The number of orders returned on Monday = $(28a-77)$

Also,

The number of delivered on Tuesday = 12% of 100a = 12a

The number of orders accepted on Tuesday = 30

The number of orders returned on Tuesday = $(12a-30)$

So,

$$(28a-77) + (12a-30) = 53$$

$$40a - 107 = 53$$

$$40a = 160$$

$$a = 4$$

@QuizByMickey

Therefore, the total number of orders delivered on all four days together = $100a = (100 \times 4) = 400$.

Monday

The total number of orders placed = 140

The number of orders delivered = 28% of 400 = 112

The number of non-delivered orders = $(140 - 112) = 28$

The number of orders accepted = 77

The number of orders returned = $(112 - 77) = 35$

Tuesday

The total number of orders placed = 64

The number of orders delivered = 12% of 400 = 48

The number of non-delivered orders = $(64 - 48) = 16$

The number of orders accepted = 30

The number of orders returned = $(48 - 30) = 18$

Now, let the number of orders accepted on Wednesday and Thursday be $8b$ and $5b$ respectively.

Wednesday

The total number of orders placed = 200

The number of orders delivered = 35% of 400 = 140

The number of non-delivered orders = $(200 - 140) = 60$

The number of orders accepted = $8b$

The number of orders returned = $(140 - 8b)$

Thursday

The total number of orders placed = 120

The number of orders delivered = 25% of 400 = 100

The number of non-delivered orders = $(120 - 100) = 20$

The number of orders accepted = $5b$

The number of orders returned = $(100 - 5b)$

So,

$$(100 - 5b) - (140 - 8b) = 5$$

$$95 - 5b = 140 - 8b$$

$$3b = 45$$

$$b = 15$$

@QuizByMickey

Therefore,

The number of orders accepted on Wednesday = $8b = (8 \times 15) = 120$

The number of orders accepted on Thursday = $5b = (5 \times 15) = 75$

The number of orders returned on Wednesday = $(140 - 8b) = (140 - 120) = 20$

The number of orders returned on Thursday = $(100 - 5b) = (100 - 75) = 25$

Day	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of orders accepted	The number of orders returned
Monday	140	112	28	77	35
Tuesday	64	48	16	30	18
Wednesday	200	140	60	120	20
Thursday	120	100	20	75	25
Total	524	400	124	302	98

Answer: D

Monday

The total number of orders delivered = 112

The total number of grocery items delivered = $1/4$ of 112 = 28

The total number of electronic items delivered = $(112 - 28) = 84$

The total number of orders returned = 35

The number of electronic items returned = $3/5$ of 35 = 21

Now, the total number of orders accepted = 77

The number of electronic items accepted = $(84 - 21) = 63$

Therefore, the required percentage = $(63/77) \times 100 = 81.81\%$.

Hence, the correct answer is option D.

@QuizByMickey

9. Questions

Let the total number of orders delivered on all four days together be $100a$.

Now,

The number of delivered on Monday = 28% of $100a = 28a$

The number of orders accepted on Monday = $(30 + 47) = 77$

The number of orders returned on Monday = $(28a - 77)$

Also,

The number of delivered on Tuesday = 12% of $100a = 12a$

The number of orders accepted on Tuesday = 30

The number of orders returned on Tuesday = $(12a-30)$

So,

$$(28a-77) + (12a-30) = 53$$

$$40a - 107 = 53$$

$$40a = 160$$

$$a = 4$$

Therefore, the total number of orders delivered on all four days together = $100a = (100 \times 4) = 400$.

Monday

The total number of orders placed = 140

The number of orders delivered = 28% of 400 = 112

The number of non-delivered orders = $(140-112) = 28$

The number of orders accepted = 77

The number of orders returned = $(112-77) = 35$

Tuesday

The total number of orders placed = 64

The number of orders delivered = 12% of 400 = 48

The number of non-delivered orders = $(64-48) = 16$

The number of orders accepted = 30

The number of orders returned = $(48-30) = 18$

Now, let the number of orders accepted on Wednesday and Thursday be $8b$ and $5b$ respectively.

Wednesday

The total number of orders placed = 200

The number of orders delivered = 35% of 400 = 140

The number of non-delivered orders = $(200-140) = 60$

The number of orders accepted = $8b$

The number of orders returned = $(140-8b)$

Thursday

The total number of orders placed = 120

The number of orders delivered = 25% of 400 = 100

@QuizByMickey

The number of non-delivered orders = $(120-100) = 20$

The number of orders accepted = $5b$

The number of orders returned = $(100-5b)$

So,

$$(100-5b) - (140-8b) = 5$$

$$95 - 5b = 140 - 8b$$

$$3b = 45$$

$$b = 15$$

Therefore,

The number of orders accepted on Wednesday = $8b = (8 \times 15) = 120$

The number of orders accepted on Thursday = $5b = (5 \times 15) = 75$

The number of orders returned on Wednesday = $(140-8b) = (140-120) = 20$

The number of orders returned on Thursday = $(100-5b) = (100-75) = 25$

Day	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of orders accepted	The number of orders returned
Monday	140	112	28	77	35
Tuesday	64	48	16	30	18
Wednesday	200	140	60	120	20
Thursday	120	100	20	75	25
Total	524	400	124	302	98

Answer: A

Tuesday

The total number of orders accepted = 30

Let the number of orders sold be m .

The number of orders unsold = $(30-m)$

So, the total cost of sold orders = $(200 \times m) = 200m$

And, the total cost of unsold orders = $\{200 \times (30-m)\} = (6000-200m)$

Now,

$$200m - (6000-200m) = 2000$$

$$400m - 6000 = 2000$$

@QuizByMickey

$$400m = 8000$$

$$m = 20$$

So, the number of orders sold = $m = 20$

And, the number of orders unsold = $(30-20) = 10$

Therefore, the required ratio = $20:10 = 2:1$.

Hence, the correct answer is option A.

10. Questions

Let the total number of orders delivered on all four days together be $100a$.

Now,

The number of delivered on Monday = 28% of $100a = 28a$

The number of orders accepted on Monday = $(30+47) = 77$

The number of orders returned on Monday = $(28a-77)$

Also,

The number of delivered on Tuesday = 12% of $100a = 12a$

The number of orders accepted on Tuesday = 30

The number of orders returned on Tuesday = $(12a-30)$

So,

$$(28a-77) + (12a-30) = 53$$

$$40a - 107 = 53$$

$$40a = 160$$

$$a = 4$$

Therefore, the total number of orders delivered on all four days together = $100a = (100 \times 4) = 400$.

Monday

The total number of orders placed = 140

The number of orders delivered = 28% of $400 = 112$

The number of non-delivered orders = $(140-112) = 28$

The number of orders accepted = 77

The number of orders returned = $(112-77) = 35$

Tuesday

The total number of orders placed = 64

The number of orders delivered = 12% of $400 = 48$

@QuizByMickey

The number of non-delivered orders = $(64-48) = 16$

The number of orders accepted = 30

The number of orders returned = $(48-30) = 18$

Now, let the number of orders accepted on Wednesday and Thursday be $8b$ and $5b$ respectively.

Wednesday

The total number of orders placed = 200

The number of orders delivered = 35% of 400 = 140

The number of non-delivered orders = $(200-140) = 60$

The number of orders accepted = $8b$

The number of orders returned = $(140-8b)$

Thursday

The total number of orders placed = 120

The number of orders delivered = 25% of 400 = 100

The number of non-delivered orders = $(120-100) = 20$

The number of orders accepted = $5b$

The number of orders returned = $(100-5b)$

So,

$$(100-5b) - (140-8b) = 5$$

$$95 - 5b = 140 - 8b$$

$$3b = 45$$

$$b = 15$$

Therefore,

The number of orders accepted on Wednesday = $8b = (8 \times 15) = 120$

The number of orders accepted on Thursday = $5b = (5 \times 15) = 75$

The number of orders returned on Wednesday = $(140-8b) = (140-120) = 20$

The number of orders returned on Thursday = $(100-5b) = (100-75) = 25$

@QuizByMickey

Day	The total number of orders placed	The number of orders delivered	The number of non-delivered orders	The number of orders accepted	The number of orders returned
Monday	140	112	28	77	35
Tuesday	64	48	16	30	18
Wednesday	200	140	60	120	20
Thursday	120	100	20	75	25
Total	524	400	124	302	98

Answer: D

Saturday

The ratio of the number of orders delivered and non-delivered orders = 5:1. (Note: 16.67% = 1/6)

The number of orders delivered = $(100 \times 2) = 200$

So, the number of non-delivered orders = $\{(200/5) \times 1\} = 40$

Therefore, the total number of orders placed on Saturday = $(200+40) = 240$

Hence, the correct answer is option D.

11. Questions

2022

The total number of furniture (tables + chairs) manufactured = 500

The total number of chairs manufactured = $2/5$ of 500 = 200

The total number of tables manufactured = $(500-200) = 300$

The ratio of the number of tables unsold and chairs sold = 40:100 = 2:5. Let it be 2a and 5a respectively.

Also, let the number of tables sold and chairs unsold be 24b and 5b respectively.

Now,

$$2a + 24b = 300$$

$$a + 12b = 150 \dots\dots\dots(I)$$

And,

$$5a + 5b = 200$$

$$a + b = 40 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$11b = 110$$

$$b = 10$$

@QuizByMickey

Now, put the value of b in equation (I), we get,

$$a + 12b = 150$$

$$a + (12 \times 10) = 150$$

$$a = 150 - 120$$

$$a = 30$$

Therefore,

$$\text{The number of tables sold} = 24b = (24 \times 10) = 240$$

$$\text{The number of tables unsold} = 2a = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = 5a = (5 \times 30) = 150$$

$$\text{The number of chairs unsold} = 5b = (5 \times 10) = 50$$

2023

$$\text{The total number of tables manufactured} = (300 - 20) = 280$$

$$\text{The total number of chairs manufactured} = \{(280/70) \times 30\} = 120$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (280 + 120) = 400$$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2m$ and $5m$ respectively.

Also, the ratio of the number of tables sold and chairs unsold = $12:1$. Let it be $12n$ and n respectively.

Now,

$$2m + 12n = 280$$

$$m + 6n = 140 \dots \dots \dots \text{(III)}$$

And,

$$5m + n = 120 \dots \dots \dots \text{(IV)}$$

By solving equation (III) and (IV), we get,

$$5m + 30n = 700$$

$$5m + n = 120$$

$$29n = 580$$

$$n = 20$$

Now, put the value of n in equation (III), we get,

$$m + 6n = 140$$

$$m + (6 \times 20) = 140$$

$$m = 140 - 120$$

$$m = 20$$

@QuizByMickey

Therefore,

$$\text{The number of tables sold} = 12n = (12 \times 20) = 240$$

$$\text{The number of tables unsold} = 2m = (2 \times 20) = 40$$

$$\text{The number of chairs sold} = 5m = (5 \times 20) = 100$$

$$\text{The number of chairs unsold} = n = 20$$

2024

$$\text{The total number of chairs manufactured} = 50$$

$$\text{The total number of tables manufactured} = (50 \times 3) = 150$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (150 + 50) = 200$$

Now, let the number of tables sold and chairs unsold be $9p$ and $2p$ respectively.

Also, the ratio of the number of chairs sold and tables unsold = $1:2$. Let it be q and $2q$ respectively.

So,

$$9p + 2q = 150 \dots\dots\dots (V)$$

And,

$$2p + q = 50 \dots\dots\dots (VI)$$

By solving equations (V) and (VI), we get,

$$18p + 4q = 300$$

$$18p + 9q = 450$$

$$5q = 150$$

$$q = 30$$

Now, put the value of q in equation (V), we get,

$$9p + 2q = 150$$

$$9p + (2 \times 30) = 150$$

$$9p + 60 = 150$$

$$9p = 90$$

$$p = 10$$

Therefore,

$$\text{The number of tables sold} = 9p = (9 \times 10) = 90$$

$$\text{The number of tables unsold} = 2q = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = q = 30$$

$$\text{The number of chairs unsold} = 2p = (2 \times 10) = 20$$

@QuizByMickey

Tables

Year	The total number of tables manufactured	The number of tables sold	The number of tables unsold
2022	300	240	60
2023	280	240	40
2024	150	90	60
Total	730	570	160

Chairs

Year	The total number of chairs manufactured	The number of chairs sold	The number of chairs unsold
2022	200	150	50
2023	120	100	20
2024	50	30	20
Total	370	280	90

Answer: D

Now, according to the question,

2022

The total number of tables sold = 240 (all non-defective)

So, the number of non-defective tables unsold = $\frac{1}{8}$ of 240 = 30

So, the total number of non-defective tables = $(240+30) = 270$

And, the total number of tables manufactured = 300

So, the number of defective tables = $(300-270) = 30$

Therefore, $X = \{(30/300) \times 100\} = 10$.

Hence, the correct answer is option D.

12. Questions

2022

The total number of furniture (tables + chairs) manufactured = 500

The total number of chairs manufactured = $\frac{2}{5}$ of 500 = 200

@QuizByMickey

The total number of tables manufactured = $(500-200) = 300$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2a$ and $5a$ respectively.

Also, let the number of tables sold and chairs unsold be $24b$ and $5b$ respectively.

Now,

$$2a + 24b = 300$$

$$a + 12b = 150 \dots\dots\dots (I)$$

And,

$$5a + 5b = 200$$

$$a + b = 40 \dots\dots\dots (II)$$

By solving equations (I) and (II), we get,

$$11b = 110$$

$$b = 10$$

Now, put the value of b in equation (I), we get,

$$a + 12b = 150$$

$$a + (12 \times 10) = 150$$

$$a = 150 - 120$$

$$a = 30$$

Therefore,

$$\text{The number of tables sold} = 24b = (24 \times 10) = 240$$

$$\text{The number of tables unsold} = 2a = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = 5a = (5 \times 30) = 150$$

$$\text{The number of chairs unsold} = 5b = (5 \times 10) = 50$$

2023

$$\text{The total number of tables manufactured} = (300-20) = 280$$

$$\text{The total number of chairs manufactured} = \{(280/70) \times 30\} = 120$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (280+120) = 400$$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2m$ and $5m$ respectively.

Also, the ratio of the number of tables sold and chairs unsold = $12:1$. Let it be $12n$ and n respectively.

Now,

$$2m + 12n = 280$$

$$m + 6n = 140 \dots\dots\dots (III)$$

@QuizByMickey

And,

$$5m + n = 120 \dots\dots\dots(IV)$$

By solving equation (III) and (IV), we get,

$$5m + 30n = 700$$

$$5m + n = 120$$

$$29n = 580$$

$$n = 20$$

Now, put the value of n in equation (III), we get,

$$m + 6n = 140$$

$$m + (6 \times 20) = 140$$

$$m = 140 - 120$$

$$m = 20$$

Therefore,

$$\text{The number of tables sold} = 12n = (12 \times 20) = 240$$

$$\text{The number of tables unsold} = 2m = (2 \times 20) = 40$$

$$\text{The number of chairs sold} = 5m = (5 \times 20) = 100$$

$$\text{The number of chairs unsold} = n = 20$$

2024

$$\text{The total number of chairs manufactured} = 50$$

$$\text{The total number of tables manufactured} = (50 \times 3) = 150$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (150 + 50) = 200$$

Now, let the number of tables sold and chairs unsold be $9p$ and $2p$ respectively.

Also, the ratio of the number of chairs sold and tables unsold = 1:2. Let it be q and $2q$ respectively.

So,

$$9p + 2q = 150 \dots\dots\dots(V)$$

And,

$$2p + q = 50 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$18p + 4q = 300$$

$$18p + 9q = 450$$

$$5q = 150$$

@QuizByMickey

$$q = 30$$

Now, put the value of q in equation (V), we get,

$$9p + 2q = 150$$

$$9p + (2 \times 30) = 150$$

$$9p + 60 = 150$$

$$9p = 90$$

$$p = 10$$

Therefore,

$$\text{The number of tables sold} = 9p = (9 \times 10) = 90$$

$$\text{The number of tables unsold} = 2q = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = q = 30$$

$$\text{The number of chairs unsold} = 2p = (2 \times 10) = 20$$

Tables

Year	The total number of tables manufactured	The number of tables sold	The number of tables unsold
2022	300	240	60
2023	280	240	40
2024	150	90	60
Total	730	570	160

Chairs

Year	The total number of chairs manufactured	The number of chairs sold	The number of chairs unsold
2022	200	150	50
2023	120	100	20
2024	50	30	20
Total	370	280	90

@QuizByMickey

Answer: A

2023

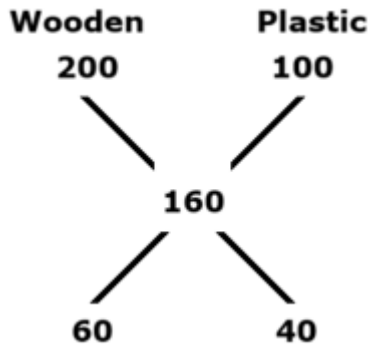
The price of each wooden chair = Rs. 200

The price of each plastic chair = Rs. 100

The total number of chairs (wooden + plastic) sold = 100

So, the average price of each chair (wooden + plastic) = $(16000/100)$ = Rs. 160

Now, using allegation method,



So, the ratio of the number of wooden and plastic chairs sold = 60:40 = 3:2.

Therefore, the number of wooden chairs sold by the shop in 2023 = $3/5$ of 100 = 60.

Hence, the correct answer is option A.

13. Questions

2022

The total number of furniture (tables + chairs) manufactured = 500

The total number of chairs manufactured = $2/5$ of 500 = 200

The total number of tables manufactured = $(500-200)$ = 300

The ratio of the number of tables unsold and chairs sold = 40:100 = 2:5. Let it be 2a and 5a respectively.

Also, let the number of tables sold and chairs unsold be 24b and 5b respectively.

Now,

$$2a + 24b = 300$$

$$a + 12b = 150 \dots\dots\dots(I)$$

And,

$$5a + 5b = 200$$

$$a + b = 40 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$11b = 110$$

$$b = 10$$

Now, put the value of b in equation (I), we get,

@QuizByMickey

$$a + 12b = 150$$

$$a + (12 \times 10) = 150$$

$$a = 150 - 120$$

$$a = 30$$

Therefore,

$$\text{The number of tables sold} = 24b = (24 \times 10) = 240$$

$$\text{The number of tables unsold} = 2a = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = 5a = (5 \times 30) = 150$$

$$\text{The number of chairs unsold} = 5b = (5 \times 10) = 50$$

2023

$$\text{The total number of tables manufactured} = (300 - 20) = 280$$

$$\text{The total number of chairs manufactured} = \{(280/70) \times 30\} = 120$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (280 + 120) = 400$$

The ratio of the number of tables unsold and chairs sold = 40:100 = 2:5. Let it be $2m$ and $5m$ respectively.

Also, the ratio of the number of tables sold and chairs unsold = 12:1. Let it be $12n$ and n respectively.

Now,

$$2m + 12n = 280$$

$$m + 6n = 140 \dots\dots\dots\text{(III)}$$

And,

$$5m + n = 120 \dots\dots\dots\text{(IV)}$$

By solving equation (III) and (IV), we get,

$$5m + 30n = 700$$

$$5m + n = 120$$

$$29n = 580$$

$$n = 20$$

Now, put the value of n in equation (III), we get,

$$m + 6n = 140$$

$$m + (6 \times 20) = 140$$

$$m = 140 - 120$$

$$m = 20$$

Therefore,

@QuizByMickey

The number of tables sold = $12n = (12 \times 20) = 240$

The number of tables unsold = $2m = (2 \times 20) = 40$

The number of chairs sold = $5m = (5 \times 20) = 100$

The number of chairs unsold = $n = 20$

2024

The total number of chairs manufactured = 50

The total number of tables manufactured = $(50 \times 3) = 150$

The total number of furniture (tables + chairs) manufactured = $(150 + 50) = 200$

Now, let the number of tables sold and chairs unsold be $9p$ and $2p$ respectively.

Also, the ratio of the number of chairs sold and tables unsold = $1:2$. Let it be q and $2q$ respectively.

So,

$$9p + 2q = 150 \dots\dots\dots (V)$$

And,

$$2p + q = 50 \dots\dots\dots (VI)$$

By solving equations (V) and (VI), we get,

$$18p + 4q = 300$$

$$18p + 9q = 450$$

$$5q = 150$$

$$q = 30$$

Now, put the value of q in equation (V), we get,

$$9p + 2q = 150$$

$$9p + (2 \times 30) = 150$$

$$9p + 60 = 150$$

$$9p = 90$$

$$p = 10$$

Therefore,

$$\text{The number of tables sold} = 9p = (9 \times 10) = 90$$

$$\text{The number of tables unsold} = 2q = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = q = 30$$

$$\text{The number of chairs unsold} = 2p = (2 \times 10) = 20$$

Tables

@QuizByMickey

Year	The total number of tables manufactured	The number of tables sold	The number of tables unsold
2022	300	240	60
2023	280	240	40
2024	150	90	60
Total	730	570	160

Chairs

Year	The total number of chairs manufactured	The number of chairs sold	The number of chairs unsold
2022	200	150	50
2023	120	100	20
2024	50	30	20
Total	370	280	90

Answer: B

The number of furniture (tables + chairs) sold in 2022 = $(240+150) = 390$

2021

The number of furniture sold = $(390+90) = 480$

The number of furniture sold to males = 360

So, the number of furniture sold to females = $(480-360) = 120$

And, the number of furniture unsold = 120

Therefore, the total number of furniture manufactured in 2021 = $(480+120) = 600$.

Hence, the correct answer is option B.

14. Questions

2022

The total number of furniture (tables + chairs) manufactured = 500

The total number of chairs manufactured = $\frac{2}{5}$ of 500 = 200

The total number of tables manufactured = $(500-200) = 300$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2a$ and $5a$ respectively.

Also, let the number of tables sold and chairs unsold be $24b$ and $5b$ respectively.

Now,

$$2a + 24b = 300$$

$$a + 12b = 150 \dots\dots\dots (I)$$

And,

$$5a + 5b = 200$$

$$a + b = 40 \dots\dots\dots (II)$$

By solving equations (I) and (II), we get,

$$11b = 110$$

$$b = 10$$

Now, put the value of b in equation (I), we get,

$$a + 12b = 150$$

$$a + (12 \times 10) = 150$$

$$a = 150 - 120$$

$$a = 30$$

Therefore,

$$\text{The number of tables sold} = 24b = (24 \times 10) = 240$$

$$\text{The number of tables unsold} = 2a = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = 5a = (5 \times 30) = 150$$

$$\text{The number of chairs unsold} = 5b = (5 \times 10) = 50$$

2023

$$\text{The total number of tables manufactured} = (300 - 20) = 280$$

$$\text{The total number of chairs manufactured} = \{(280/70) \times 30\} = 120$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (280 + 120) = 400$$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2m$ and $5m$ respectively.

Also, the ratio of the number of tables sold and chairs unsold = $12:1$. Let it be $12n$ and n respectively.

Now,

$$2m + 12n = 280$$

$$m + 6n = 140 \dots\dots\dots (III)$$

And,

$$5m + n = 120 \dots\dots\dots (IV)$$

By solving equation (III) and (IV), we get,

$$5m + 30n = 700$$

$$5m + n = 120$$

$$29n = 580$$

$$n = 20$$

Now, put the value of n in equation (III), we get,

$$m + 6n = 140$$

$$m + (6 \times 20) = 140$$

$$m = 140 - 120$$

$$m = 20$$

Therefore,

$$\text{The number of tables sold} = 12n = (12 \times 20) = 240$$

$$\text{The number of tables unsold} = 2m = (2 \times 20) = 40$$

$$\text{The number of chairs sold} = 5m = (5 \times 20) = 100$$

$$\text{The number of chairs unsold} = n = 20$$

2024

$$\text{The total number of chairs manufactured} = 50$$

$$\text{The total number of tables manufactured} = (50 \times 3) = 150$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (150 + 50) = 200$$

Now, let the number of tables sold and chairs unsold be $9p$ and $2p$ respectively.

Also, the ratio of the number of chairs sold and tables unsold = $1:2$. Let it be q and $2q$ respectively.

So,

$$9p + 2q = 150 \dots\dots\dots(V)$$

And,

$$2p + q = 50 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$18p + 4q = 300$$

$$18p + 9q = 450$$

$$5q = 150$$

$$q = 30$$

Now, put the value of q in equation (V), we get,

@QuizByMickey

$$9p + 2q = 150$$

$$9p + (2 \times 30) = 150$$

$$9p + 60 = 150$$

$$9p = 90$$

$$p = 10$$

Therefore,

$$\text{The number of tables sold} = 9p = (9 \times 10) = 90$$

$$\text{The number of tables unsold} = 2q = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = q = 30$$

$$\text{The number of chairs unsold} = 2p = (2 \times 10) = 20$$

Tables

Year	The total number of tables manufactured	The number of tables sold	The number of tables unsold
2022	300	240	60
2023	280	240	40
2024	150	90	60
Total	730	570	160

Chairs

Year	The total number of chairs manufactured	The number of chairs sold	The number of chairs unsold
2022	200	150	50
2023	120	100	20
2024	50	30	20
Total	370	280	90

@QuizByMickey

Answer: C

2024

The number of furniture unsold initially = 80

So, $Y\%$ of 80 = $0.80Y$

The ratio of the number of furniture sold and unsold will be 7:3

Now, the total number of furniture = 200

And, the number of furniture sold = $200 \times (7/10) = 140$

The number of furniture unsold = $200 \times (3/10) = 60$

So,

The number of furniture unsold are further sold = $80 - 60 = 20$

$0.80Y = 20$

$Y = 25$

Therefore, $2Y = (2 \times 25) = 50$.

Hence, the correct answer is option C.

15. Questions

2022

The total number of furniture (tables + chairs) manufactured = 500

The total number of chairs manufactured = $2/5$ of $500 = 200$

The total number of tables manufactured = $(500 - 200) = 300$

The ratio of the number of tables unsold and chairs sold = $40:100 = 2:5$. Let it be $2a$ and $5a$ respectively.

Also, let the number of tables sold and chairs unsold be $24b$ and $5b$ respectively.

Now,

$$2a + 24b = 300$$

$$a + 12b = 150 \dots \dots \dots (I)$$

And,

$$5a + 5b = 200$$

$$a + b = 40 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$11b = 110$$

$$b = 10$$

Now, put the value of b in equation (I), we get,

$$a + 12b = 150$$

$$a + (12 \times 10) = 150$$

$$a = 150 - 120$$

$$a = 30$$

@QuizByMickey

Therefore,

$$\text{The number of tables sold} = 24b = (24 \times 10) = 240$$

$$\text{The number of tables unsold} = 2a = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = 5a = (5 \times 30) = 150$$

$$\text{The number of chairs unsold} = 5b = (5 \times 10) = 50$$

2023

$$\text{The total number of tables manufactured} = (300 - 20) = 280$$

$$\text{The total number of chairs manufactured} = \{(280/70) \times 30\} = 120$$

$$\text{The total number of furniture (tables + chairs) manufactured} = (280 + 120) = 400$$

$$\text{The ratio of the number of tables unsold and chairs sold} = 40:100 = 2:5. \text{ Let it be } 2m \text{ and } 5m \text{ respectively.}$$

$$\text{Also, the ratio of the number of tables sold and chairs unsold} = 12:1. \text{ Let it be } 12n \text{ and } n \text{ respectively.}$$

Now,

$$2m + 12n = 280$$

$$m + 6n = 140 \dots\dots\dots \text{(III)}$$

And,

$$5m + n = 120 \dots\dots\dots \text{(IV)}$$

By solving equation (III) and (IV), we get,

$$5m + 30n = 700$$

$$5m + n = 120$$

$$29n = 580$$

$$n = 20$$

Now, put the value of n in equation (III), we get,

$$m + 6n = 140$$

$$m + (6 \times 20) = 140$$

$$m = 140 - 120$$

$$m = 20$$

Therefore,

$$\text{The number of tables sold} = 12n = (12 \times 20) = 240$$

$$\text{The number of tables unsold} = 2m = (2 \times 20) = 40$$

$$\text{The number of chairs sold} = 5m = (5 \times 20) = 100$$

$$\text{The number of chairs unsold} = n = 20$$

@QuizByMickey

2024

The total number of chairs manufactured = 50

The total number of tables manufactured = $(50 \times 3) = 150$

The total number of furniture (tables + chairs) manufactured = $(150 + 50) = 200$

Now, let the number of tables sold and chairs unsold be $9p$ and $2p$ respectively.

Also, the ratio of the number of chairs sold and tables unsold = $1:2$. Let it be q and $2q$ respectively.

So,

$$9p + 2q = 150 \dots\dots\dots (V)$$

And,

$$2p + q = 50 \dots\dots\dots (VI)$$

By solving equations (V) and (VI), we get,

$$18p + 4q = 300$$

$$18p + 9q = 450$$

$$5q = 150$$

$$q = 30$$

Now, put the value of q in equation (V), we get,

$$9p + 2q = 150$$

$$9p + (2 \times 30) = 150$$

$$9p + 60 = 150$$

$$9p = 90$$

$$p = 10$$

Therefore,

$$\text{The number of tables sold} = 9p = (9 \times 10) = 90$$

$$\text{The number of tables unsold} = 2q = (2 \times 30) = 60$$

$$\text{The number of chairs sold} = q = 30$$

$$\text{The number of chairs unsold} = 2p = (2 \times 10) = 20$$

Tables

@QuizByMickey

Year	The total number of tables manufactured	The number of tables sold	The number of tables unsold
2022	300	240	60
2023	280	240	40
2024	150	90	60
Total	730	570	160

Chairs

Year	The total number of chairs manufactured	The number of chairs sold	The number of chairs unsold
2022	200	150	50
2023	120	100	20
2024	50	30	20
Total	370	280	90

Answer: B

2022

The total number of chairs manufactured = 200

Now, after some chairs returned,

The number of chairs unsold = $150 - 85 = 65$

And, the number of chairs sold = $(200 - 65) = 135$

The initial number of chairs sold = 150

Therefore, $? = \left[\frac{(150 - 135)}{150} \times 100 \right] = \left[\frac{(15)}{150} \times 100 \right] = 10$.

Hence, the correct answer is option B.

@QuizByMickey

16. Questions

Answer: D

Series I: 33, 11, 66, __[P]__, 132, 44

1st term = 33

2nd term = $(33/3) = 11$

$$3^{\text{rd}} \text{ term} = (11 \times 6) = 66$$

$$4^{\text{th}} \text{ term} = (66/3) = 22 \text{ (P)}$$

$$5^{\text{th}} \text{ term} = (22 \times 6) = 132$$

$$6^{\text{th}} \text{ term} = (132/3) = 44$$

So, the missing term = P = 22

Series II: 14, 34, 9, 39, 4, __[Q]__

$$1^{\text{st}} \text{ term} = 14$$

$$2^{\text{nd}} \text{ term} = (14 + 20) = 34$$

$$3^{\text{rd}} \text{ term} = (34 - 25) = 9$$

$$4^{\text{th}} \text{ term} = (9 + 30) = 39$$

$$5^{\text{th}} \text{ term} = (39 - 35) = 4$$

$$6^{\text{th}} \text{ term} = (4 + 40) = 44 \text{ (Q)}$$

So, the missing term = Q = 44

$$\text{I: } 2P = Q$$

$$(2 \times 22) = 44$$

$$Q = 44$$

So, I is correct.

$$\text{II: } (Q - 22) = P$$

$$(44 - 22) = 22$$

$$P = 22$$

So, II is correct.

$$\text{III: } (Q/P) = 2$$

$$(44/22) = 2$$

So, III is correct.

Therefore, all I, II, and III follow.

Hence, the correct answer is option D.

17. Questions

Answer: D

Series I: 3, 5, 13, 48, 241, 1441

@QuizByMickey

$$1^{\text{st}} \text{ term} = 3$$

$$2^{\text{nd}} \text{ term} = (3 \times 2 - 1) = (6 - 1) = 5$$

$$3^{\text{rd}} \text{ term} = (5 \times 3 - 2) = (15 - 2) = 13$$

$$4^{\text{th}} \text{ term} = (13 \times 4 - 3) = (52 - 3) = 49 \text{ (48 is wrong)}$$

$$5^{\text{th}} \text{ term} = (49 \times 5 - 4) = (245 - 4) = 241$$

$$6^{\text{th}} \text{ term} = (241 \times 6 - 5) = (1446 - 5) = 1441$$

So, the wrong term = 48

And, the correct term = K = 49

Series II: 8, 15, 43, __[M]__, __[N]__, 5041

$$1^{\text{st}} \text{ term} = 8$$

$$2^{\text{nd}} \text{ term} = (8 \times 2 - 1) = (16 - 1) = 15$$

$$3^{\text{rd}} \text{ term} = (15 \times 3 - 2) = (45 - 2) = 43$$

$$4^{\text{th}} \text{ term} = (43 \times 4 - 3) = (172 - 3) = 169 \text{ (M)}$$

$$5^{\text{th}} \text{ term} = (169 \times 5 - 4) = (845 - 4) = 841 \text{ (N)}$$

$$6^{\text{th}} \text{ term} = (841 \times 6 - 5) = (5046 - 5) = 5041$$

So, M = 169

And, N = 841

Now,

49 is a perfect square of 7

169 is a perfect square of 13

841 is a perfect square of 29

Therefore, all K, M, and N are perfect squares.

Hence, the correct answer is option D.

18. Questions

Answer: C

Series I: 10, 14, 23, 39, 97, 218

$$1^{\text{st}} \text{ term} = 10$$

$$2^{\text{nd}} \text{ term} = (10 + 2^2) = (10 + 4) = 14$$

@QuizByMickey

$$3^{\text{rd}} \text{ term} = (14+3^2) = (14+9) = 23$$

$$4^{\text{th}} \text{ term} = (23+5^2) = (23+25) = 48 \text{ (39 is wrong)}$$

$$5^{\text{th}} \text{ term} = (48+7^2) = (48+49) = 97$$

$$6^{\text{th}} \text{ term} = (97+11^2) = (97+121) = 218$$

So, the wrong term = 39

And, the correct term = $X = 48$

Series II: 30, 34, 43, 68, 104, 238

$$1^{\text{st}} \text{ term} = 30$$

$$2^{\text{nd}} \text{ term} = (30+2^2) = (30+4) = 34$$

$$3^{\text{rd}} \text{ term} = (34+3^2) = (34+9) = 43$$

$$4^{\text{th}} \text{ term} = (43+5^2) = (43+25) = 68$$

$$5^{\text{th}} \text{ term} = (68+7^2) = (68+49) = 117 \text{ (104 is wrong)}$$

$$6^{\text{th}} \text{ term} = (117+11^2) = (117+121) = 238$$

So, the wrong term = 104

And, the correct term = $Y = 117$

I: Neither 48 nor 117 is exactly divisible by 5.

So, I is definitely not true.

II: The sum of the digits of 117 = $(1+1+7) = 9$

So, II is definitely true.

III: The square of 7 = $7^2 = 49$

$X = 48$ (1 less than 49)

So, III is definitely true.

Therefore, II and III are definitely true.

Hence, the correct answer is option C.

19. Questions

Answer: B

The only possible values of $A = 3^3 = 27$

So, $A = 27$

@QuizByMickey

The two-digit number which is exactly divisible by both 5 and 11 = 55 (only possibility)

So, B = 55

The second lowest two-digit prime number = 13

So, C = $(13^2 - 5) = (169 - 5) = 164$.

Series I: 28, [A], [B], [C], 657, 3284

28, 27, 55, 164, 657, 3284

1st term = 28

2nd term = $(28 \times 1 - 1) = (28 - 1) = 27$

3rd term = $(27 \times 2 + 1) = (54 + 1) = 55$

4th term = $(55 \times 3 - 1) = (165 - 1) = 164$

5th term = $(164 \times 4 + 1) = (656 + 1) = 657$

6th term = $(657 \times 5 - 1) = (3285 - 1) = 3284$

Series II: 16, ____, ____, ____, ____, __[?]

1st term = 16

2nd term = $(16 \times 1 - 1) = (16 - 1) = 15$

3rd term = $(15 \times 2 + 1) = (30 + 1) = 31$

4th term = $(31 \times 3 - 1) = (93 - 1) = 92$

5th term = $(92 \times 4 + 1) = (368 + 1) = 369$

6th term = $(369 \times 5 - 1) = (1845 - 1) = 1844$ (?)

Therefore, ? = 1844

Hence, the correct answer is option B.

20. Questions

Answer: C

Series I: 4, 8, 2, 16, __[K]__, 32

1st term = 4

2nd term = $(4 \times 2) = 8$

3rd term = $(8/4) = 2$

@QuizByMickey

$$4^{\text{th}} \text{ term} = (2 \times 8) = 16$$

$$5^{\text{th}} \text{ term} = (16/16) = 1 \text{ (K)}$$

$$6^{\text{th}} \text{ term} = (1 \times 32) = 32$$

So, the missing term = K = 1

Series II: __[K]__, 13, 37, 86, 181, 373

1, 13, 37, 86, 181, 373

$$1^{\text{st}} \text{ term} = 1$$

$$2^{\text{nd}} \text{ term} = (1+12) = 13$$

$$3^{\text{rd}} \text{ term} = (13+24) = 37$$

$$4^{\text{th}} \text{ term} = (37+48) = 85 \text{ (86 is wrong)}$$

$$5^{\text{th}} \text{ term} = (85+96) = 181$$

$$6^{\text{th}} \text{ term} = (181+192) = 373$$

So, the wrong term = 86

And, the correct term = 85.

Hence, the correct answer is option C.

@QuizByMickey